SMART WASTEWATER TREATMENT AND REUSE SYSTEM

Sample Output — Automatic Reuse Recommendation

Sample ID: WW-DEMO-001 | Source: Demonstration wastewater | Values below are fictional examples.

1. Pre-treatment vs Post-treatment Analysis

Parameter	Pre-treatment	Post-treatment	Configured reference range	Post status
pH	7.8	7.2	6.5–8.5	In range
Turbidity (NTU)	48	3.5	0–10	In range
COD (mg/L)	320	72	0–250	In range
BOD (mg/L)	95	8	0–30	In range
TDS (mg/L)	780	650	0–500	Review
EC (µS/cm)	1650	1250	0–1500	In range
DO (mg/L)	0.8	4.2	3–100	In range

Note: these are demonstration values. Configured reference ranges are prototype defaults and are not universal regulatory limits.

2. Automatic reuse application screening

System result for this example: Landscape / ornamental plant irrigation is a potential match. The post-treatment values meet the illustrative landscape screening criteria in this sample. This is only a preliminary match, not a declaration that the water is safe.

Potential application	Automated screening	Additional note
Landscape / ornamental irrigation	Criteria met	Microbiological testing and local review required
Road cleaning / dust suppression	Criteria met	Exposure/runoff controls and local review required
Toilet flushing	Not a match in this example	Validated disinfection, microbiological testing and regulatory approval require

3. Before/after trend

Turbidity decreased from 48 to 3.5 NTU (92.7% reduction); BOD decreased from 95 to 8 mg/L (91.6% reduction); COD decreased from 320 to 72 mg/L (77.5% reduction). These calculations illustrate treatment change. For parameters such as pH and DO, lower/higher is not automatically better; interpretation depends on the intended use.

Safety limitation: The app screens only pH, turbidity, COD, BOD, TDS, electrical conductivity and dissolved oxygen. It does not measure pathogens, metals or every use-specific contaminant. Never use this sample output alone to authorize reuse. Confirm applicable standards, required laboratory testing and qualified professional approval.